Procedure – How to Perform

Step 1 – Create Project

Open PyCharm.

Create:

SoftwareTesting

Step 2 – Create File

Create:

tutorial15.py

Step 3 – Install Selenium

Open PyCharm Terminal:

pip install selenium

Step 4 – Import Selenium

```
from selenium import webdriver
from selenium.webdriver.common.by import By
```

Step 5 – Import Wait and Exceptions

```
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

from selenium.common.exceptions import (
    NoSuchElementException,
    TimeoutException,
    ElementNotInteractableException,
    StaleElementReferenceException
)
```

Step 6 – Create Browser
driver = webdriver.Chrome()
driver.maximize_window()

Step 7 – Perform Valid Login
Username: standard_user
Password: secret_sauce

Step 8 – Validate Successful Login
assert "inventory" in driver.current_url

Step 9 – Test Invalid Password
Use:
Username: standard_user
Password: wrong_password

Verify error message.

Step 10 – Test Invalid Username

Use:

Username: wrong_user

Password: secret_sauce

Verify error message.

Step 11 – Test Empty Credentials

Leave username and password blank and click Login.

Verify:

Username is required

Step 12 – Logout

Login successfully, open menu and click Logout.

Source Code

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

from selenium.common.exceptions import (
    NoSuchElementException,
    TimeoutException,
    ElementNotInteractableException,
    StaleElementReferenceException
)
```

```
def login(driver, username, password):

    driver.get(URL)

    wait = WebDriverWait(driver, 10)

    username_box = wait.until(
        EC.visibility_of_element_located(
            (By.ID, "user-name")
        )
    )

    password_box = wait.until(
        EC.visibility_of_element_located(
            (By.ID, "password")
        )
    )

    login_button = wait.until(
        EC.element_to_be_clickable(
            (By.ID, "login-button")
        )
    )

    username_box.clear()
    username_box.send_keys(username)
(
```

```
try:

    login(
        driver,
        VALID_USERNAME,
        VALID_PASSWORD
    )

    WebDriverWait(driver, 10).until(
        EC.url_contains("inventory")
    )

    assert "inventory" in driver.current_url

    products_title = WebDriverWait(
        driver, 10
    ).until(
        EC.visibility_of_element_located(
            (By.CLASS_NAME, "title")
        )
    )

    assert products_title.text == "Products"

    print("TC-01 Valid Login: PASS")
    print("Authentication Assertion: PASS")
```

```
try:

    login(
        driver,
        VALID_USERNAME,
        VALID_PASSWORD
    )

    WebDriverWait(driver, 10).until(
        EC.url_contains("inventory")
    )

    assert "inventory" in driver.current_url

    products_title = WebDriverWait(
        driver, 10
    ).until(
        EC.visibility_of_element_located(
            (By.CLASS_NAME, "title")
        )
```

```
    password_box.clear()
    password_box.send_keys(password)

    login_button.click()


# TC-01 Valid Login
def test_valid_login():

    driver = start_browser()
```

```
        except AssertionError:
            print("TC-02 Invalid Password: FAIL")

        except TimeoutException:
            print("Error message was not displayed")

        finally:
            driver.quit()


    # TC-03 Invalid Username
    def test_invalid_username():

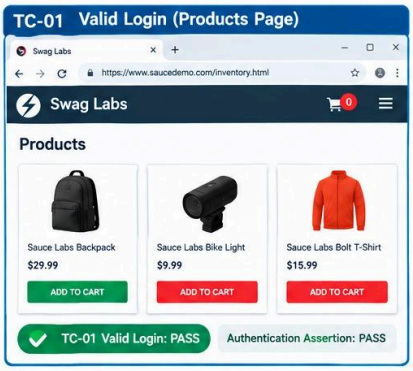
        driver = start_browser()

        try:

            login(
                driver,
                "wrong_user",
                VALID_PASSWORD
            )

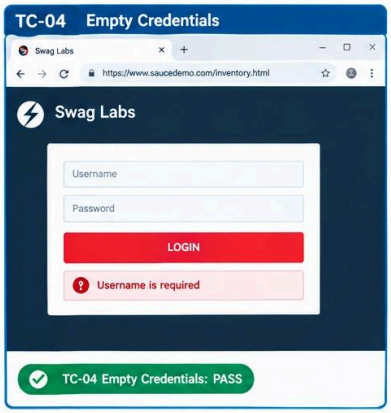
            error_message = WebDriverWait(
```

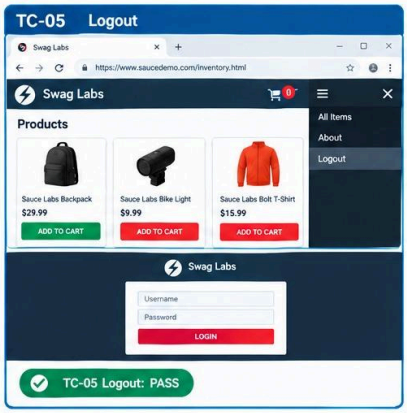
ScreenShort:

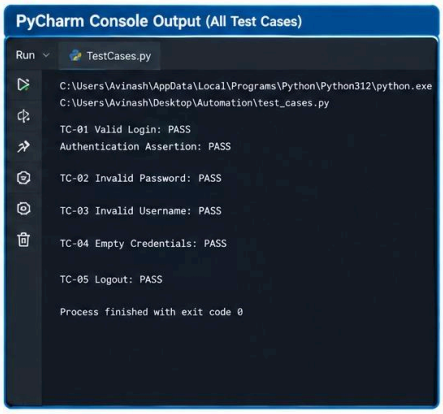

TC-01 Valid Login (Products Page)
Swag Labs
https://www.saucedemo.com/inventory.html
Swag Labs
Products
Sauce Labs Backpack
$29.99
ADD TO CART
Sauce Labs Bike Light
$9.99
ADD TO CART
Sauce Labs Bolt T-Shirt
$15.99
ADD TO CART
TC-01 Valid Login: PASS
Authentication Assertion: PASS


TC-02 Invalid Password
Swag Labs
https://www.saucedemo.com/
Swag Labs
standard_user
••••••••
LOGIN
Username and password do not match
TC-02 Invalid Password: PASS


TC-03 Invalid Username
Swag Labs
https://www.saucedemo.com/
Swag Labs
wrong_user
••••••••
LOGIN
Username and password do not match
TC-03 Invalid Username: PASS


TC-04 Empty Credentials
Swag Labs
https://www.saucedemo.com/inventory.html
Swag Labs
Username
Password
LOGIN
Username is required
TC-04 Empty Credentials: PASS


TC-05 Logout
Swag Labs
https://www.saucedemo.com/inventory.html
Swag Labs
Products
All Items
About
Logout
Sauce Labs Backpack
$29.99
ADD TO CART
Sauce Labs Bike Light
$9.99
ADD TO CART
Sauce Labs Bolt T-Shirt
$15.99
ADD TO CART
Swag Labs
Username
Password
LOGIN
TC-05 Logout: PASS


PyCharm Console Output (All Test Cases)
Run
TestCases.py
C:\Users\Avinash\AppData\Local\Programs\Python\Python312\python.exe
C:\Users\Avinash\Desktop\Automation\test_cases.py
TC-01 Valid Login: PASS
Authentication Assertion: PASS
TC-02 Invalid Password: PASS
TC-03 Invalid Username: PASS
TC-04 Empty Credentials: PASS
TC-05 Logout: PASS
Process finished with exit code 0